Control (Zone 1)

J5
Raspberry_Pi_4

Z1_DEG_A
Z1_DEG_B
Z1_INT_CAN_A
Z1_INT_CAN_B
Z1_ERR
UDP_RX
Z1_CMD_TX
Z1_CAN_TX_A
Z1_CAN_TX_B
Z1_CS_CAN_B
Z1_CS_CAN_A
Z1_MISO
Z1_MOSI
Z1_SCLK

TP1 UDP_RX
TP2 CMD_TX
TP3 CAN_TX_A
TP4 CAN_TX_B

Steering (Zone 2)

U7
STM32F401RETx

SERVO_PWM
Z2_DEG_A
Z2_DEG_B
FR_BLINK
Z2_CAN_TX_A
Z2_INT_CAN_B
Z2_CAN_RX_B
Z2_CAN_RX_A
Z2_ERR
Z2_SCLK
Z2_MISO
Z2_MOSI
Z2_CS_CAN_A
FL_BLINK
Z2_INT_CAN_A
Z2_RELAY_TX
Z2_TEST_BUTTON2

TP9 Z2_RELAY_TX
TP6 Z2_CAN_RX_B
TP7 Z2_CAN_RX_A
TP8 Z2_CAN_TX_B

Drivetrain (Zone 3)

U3
STM32F401RETx

ENC_LEFT_B
ENC_RIGHT_A
ENC_RIGHT_B
RL_BLINK
MOTOR_LEFT_PWM
Z3_INT_CAN_B
ENC_LEFT_A
MOTOR_RIGHT_PWM
MOTOR_RIGHT_DIR
Z3_SCLK
Z3_MISO
Z3_MOSI
Z3_CS_CAN_A
DIR_A
RR_BLINK
Z3_INT_CAN_A
Z3_CAN_TX_B
Z3_CAN_TX_A
Z3_CS_CAN_B
Z3_CAN_RX_B
Z3_CAN_RX_A
Z3_DEG_A
Z3_DEG_B
Z3_ERR
Z2_TEST_BUTTON
MOTOR_LEFT_DIR
PWM_SET

TP15 MOTOR_PWM_OUT
TP13 MOTOR_LEFT_DIR
TP14 PWM_SET
TP17 Z3_CAN_TX_B
TP16 Z3_CAN_TX_A
TP19 Z3_CAN_RX_B
TP18 Z3_CAN_RX_A

J6
Zone 1 (PI) Test Point Breakout

J9
Zone 2 (Front) Test Point Breakout

J10
Zone 3 (Rear) Test Point Breakout

D9 LED R R25 560 Z2_DEG_A
D10 LED R R26 560 Z2_DEG_B
D11 LED G R27 560 FL_BLINK
D12 LED G R28 560 FR_BLINK
D13 LED G R29 560 Z2_ERR

D19 LED R R35 560 Z1_DEG_A
D20 LED R R36 560 Z1_DEG_B
D23 LED G R39 560 Z1_ERR

D14 LED R R30 560 Z3_DEG_A
D15 LED R R31 560 Z3_DEG_B
D16 LED G R32 560 RL_BLINK
D17 LED G R33 560 RR_BLINK
D18 LED G R34 560 Z3_ERR

U13 MCP2515-xST
U14 SN65HVD230
Y4 CX3225SB16000H0FLJCC
C15 100nF
C16 100nF
C13 22pF
C14 22pF
R13 47K
R12 10k
Z1_MOSI
Z1_MISO
Z1_CS_CAN_B
Z1_SCLK
Z1_INT_CAN_B
CANH_B_Z1
CANL_B_Z1

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

U15 MCP2515-xST
U16 SN65HVD230
Y5 CX3225SB16000H0FLJCC
C19 100nF
C20 100nF
C17 22pF
C18 22pF
R15 47K
R14 10k
Z2_MOSI
Z2_MISO
Z2_CS_CAN_B
Z2_SCLK
Z2_INT_CAN_B
CANH_B_Z2
CANL_B_Z2

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

U17 MCP2515-xST
U18 SN65HVD230
Y6 CX3225SB16000H0FLJCC
C23 100nF
C24 100nF
C21 22pF
C22 22pF
R17 47K
R16 10k
Z3_MOSI
Z3_MISO
Z3_CS_CAN_B
Z3_SCLK
Z3_INT_CAN_B
CANH_B_Z3
CANL_B_Z3

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

U1 MCP2515-xST
U8 SN65HVD230
Y1 CX3225SB16000H0FLJCC
C4 100nF
C1 100nF
C2 22pF
C3 22pF
R2 47K
R1 10k
Z1_MOSI
Z1_MISO
Z1_CS_CAN_A
Z1_SCLK
Z1_INT_CAN_A
CANH_A_Z1
CANL_A_Z1

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

U9 MCP2515-xST
U10 SN65HVD230
Y2 CX3225SB16000H0FLJCC
C7 100nF
C8 100nF
C5 22pF
C6 22pF
R4 47K
R3 10k
Z2_MOSI
Z2_MISO
Z2_CS_CAN_A
Z2_SCLK
Z2_INT_CAN_A
CANH_A_Z2
CANL_A_Z2

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

U11 MCP2515-xST
U12 SN65HVD230
Y3 CX3225SB16000H0FLJCC
C11 100nF
C12 100nF
C9 22pF
C10 22pF
R6 47K
R5 10k
Z3_MOSI
Z3_MISO
Z3_CS_CAN_A
Z3_SCLK
Z3_INT_CAN_A
CANH_A_Z3
CANL_A_Z3

CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

JP1 JP2 JP3 JP4 JP5 JP6 JP7 JP8 JP9 JP10
CANH_A_Z1 CANH_A_Z2 CANH_A_Z3 CANH_A
CANL_A_Z1 CANL_A_Z2 CANL_A_Z3 CANL_A
R18 120 R19 120

JP11 JP12 JP13 JP14 JP15 JP16 JP17 JP18 JP19 JP20
CANH_B_Z1 CANH_B_Z2 CANH_B_Z3 CANH_B
CANL_B_Z1 CANL_B_Z2 CANL_B_Z3 CANL_B
R20 120 R21 120

SERVO_PWM
M2 Motor_Servo

U20 L298N
MOTOR_RIGHT_PWM
MOTOR_RIGHT_DIR
MOTOR_LEFT_PWM
MOTOR_LEFT_DIR

J7 Conn_01x06_Pin
ENC_RIGHT_A
ENC_LEFT_B
Right Motor

J8 Conn_01x06_Pin
ENC_LEFT_A
ENC_LEFT_B
Left Motor

Schematic is currently missing:
- Everything to do with power
- All of the oscilators and support for STM 32 processors
  - Remember to have a way to flash
- All of the current sensing
- Fun things
  - Pixels
  - Speaker

Sheet: /
File: Schematic.kicad_sch

Title: CAN Lab Scheumatic

Size: C	Date: 2026-10-22	Rev:
KiCad E.D.A. 10.0.6		Id: 1/1